

Computer Science & IT

Database Management System



Query Languages

Lecture No. 11



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Recap of Previous Lecture



- ✓ **Topic** Comparison with NULL
- ✓ **Topic** Comparison with Regular expression
- ✓ **Topic** Practice questions

Topics to be Covered



- ✓ **Topic** Introduction to relational calculus
- ✓ **Topic** Tuple relational calculus
- ✓ **Topic** Practice questions





Topic : Relational Calculus

- Relational Calculus is a non-procedural query language.
- It uses predicate calculus(First Order Logic) instead of algebra.
 $\{ \exists, \forall, \wedge, \vee, \dots \}$
- Relational calculus provides description about the query to get the result, whereas Relational Algebra gives the method to get the result.



Topic : Types of Relational Calculus

There are two types of Relational Calculus.

1. Tuple Relational Calculus: Works on tuple (or rows)
2. Domain Relational Calculus: Works on domain of attribute (column)



Topic : Syntax of TRC

Basic syntax of TRC is $\{ t \mid P(t) \}$

tuple variable

Predicate formula
over tuple variable 't'

where t is a tuple variable and $P(t)$ is a predicate formula that describes the conditions that must be satisfied by the resulting tuples.

1. It results the set of tuples t , s.t. $P(t)$ is true for t .
2. $t \in \text{Relation}$ denotes that the t is a tuple from Relation, also denoted using Relation(t).
3. If t is a tuple variable, " $t[A]$ " / " $t.A$ " specify attribute A in tuple t .
4. P is a predicate formula similar to that of the predicate calculus.

In

$\{t \mid P(t)\}$

't' is output tuple variable
ie. it is used to produce the o/p.



Topic : Equivalence between SQL, RA and TRC

SQL:- $\text{SELECT distinct * FROM Employee}$ } o/p of this query will be }
Complete Employee table }

RA:- (Employee) } O/p will be }
Complete Employee table }

TRC:- $\{ t \mid t \in \text{Employee} \}$
or
 $\{ t \mid \text{Employee}(t) \}$

$\{ t \mid \underbrace{t \in \text{Employee}}_{\text{Employee}(t)} \}$

it is our predicate formula.

this predicate formula will return true for all the tuples of Employee table

or o/p will contain all tuples of Emp table



Topic : Free and Bounded variables

- A variable which is preceded by a quantifier is called bounded variable.
 $\forall$ or $\exists$
- A variable which is not preceded by any quantifier is called free variable.
- **There is an important restriction:**
The variable 't' that appears before '|' symbol must be the only free variable in formula $P(t)$.
In other words all other variables must be bounded using quantifier



Topic : Equivalence between SQL, RA and TRC

➤ SELECT distinct ename FROM Employee

$\Rightarrow$ o/p =

o/p =

ename
-
-
-
-

tuples will be distinctinct 'ename' from Employee table

➤ $\pi_{\text{ename}}(\text{Employee})$

➤ { t.ename | t $\in$ Employee }
or
{ t[ename] | t $\in$ Employee }
or

{ t | $\exists S \in \text{Employee} (t[\text{ename}] = S[\text{ename}])$ }

or

{ t | $\exists S \in \text{Employee} (t.\text{ename} = S.\text{ename})$ }

Can be used interchangeably

{ t.ename | t $\in$ Employee }

{ t[ename] | Employee(t) }

{ t | $\exists S \in \text{Employee} (t.\text{ename} = S.\text{ename})$ }

Attribute creation w.r.t. o/p tuple 't'

$$\{ t \mid \exists S \in \text{Employee} \wedge (t.\text{ename} = S.\text{ename}) \}$$

#Q. Retrieve tuples from the relation Student where marks are more than 50
 Student(sid, name, marks)

$$(1) \rightarrow \{ t \mid \underbrace{t \in \text{Student} \wedge t.\text{marks} > 50}_{\text{it is our } P(t)} \}$$

$$(2) \{ t \mid t \in \text{Student} (t.\text{marks} > 50) \}$$

$$(3) \{ t \mid \exists S \in \text{Student} (S.\text{marks} > 50 \wedge (t.\text{sid} = S.\text{sid} \wedge t.\text{name} = S.\text{name} \wedge t.\text{marks} = S.\text{marks})) \}$$

Selection of attributes for o/p tuple 't'

#Q.

Retrieve sids of all the students from the relation S Student whose marks are more than 50.

Student(sid, name, marks)



$$\textcircled{1} \quad \{ t.\text{sid} \mid t \in \text{Student} \wedge t.\text{marks} > 50 \}$$

$$\textcircled{2} \quad \{ t \mid \exists S \in \text{Student} (S.\text{marks} > 50 \wedge (t.\text{sid} = S.\text{sid})) \}$$

#Q.

Retrieve sids and names of all the students from the relation S whose marks are more than 50.

Student(sid, name, marks)



$$\textcircled{1} \quad \{ t.\text{sid}, t.\text{name} \mid t \in \text{Student} \wedge t.\text{marks} > 50 \}$$

$$\textcircled{2} \quad \left\{ t \mid \exists S \in \text{Student} \left(S.\text{marks} > 50 \wedge (t.\text{sid} = S.\text{sid} \wedge t.\text{name} = S.\text{name}) \right) \right\}$$

#Q. Consider the following relational schema:

✓ Suppliers(sid, sname, city)

Parts(pid, pname, color)

✓ Catalog(sid, pid, cost)

Find sid of all the suppliers who have supplied part with pid=P1

$$\textcircled{1} \quad \{ t.sid \mid t \in \text{Catalog} \wedge t.pid = P_1 \}$$

$$\textcircled{2} \quad \{ t \mid \exists c \in \text{Catalog} (c.pid = P_1 \wedge (t.sid = c.sid)) \}$$

#Q.

Consider the following relational schema:

Suppliers(sid, sname, city)

Parts(pid, pname, color)

Catalog(sid, pid, cost)

Find sids of all the suppliers who have supplied some 'red' ^{color} ^{parts}

$$\textcircled{1} \{ t.\text{sid} \mid t \in \text{Catalog} \wedge (\exists \bar{P} \in \text{Parts} (t.\text{pid} = \bar{P}.\text{pid} \wedge \bar{P}.\text{Color} = \text{'Red'})) \}$$

it is predicate formula $P(t)$

$$\textcircled{2} \{ t \mid \exists C \in \text{Catalog} (\exists P \in \text{Parts} (C.\text{pid} = P.\text{pid} \wedge P.\text{Color} = \text{'Red'}) \wedge (t.\text{sid} = C.\text{sid})) \}$$

$$\textcircled{3} \{ t \mid \exists C \in \text{Catalog} (t.\text{sid} = C.\text{sid} \wedge \exists P \in \text{Parts} (C.\text{pid} = P.\text{pid} \wedge P.\text{Color} = \text{'Red'})) \}$$

#Q. Consider the following relational schema:

S = Suppliers(sid, sname, city)

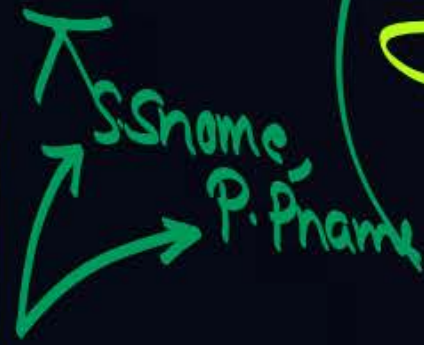
P = Parts(pid, pname, color)

C = Catalog(sid, pid, cost)

Find Sname of all the suppliers who have supplied some 'red' color parts along with the names of those parts.

Sid	Sname
S ₁	A
S ₂	B
S ₃	C

Pid	Pname	Color
P ₁	X	Red
P ₂	Y	Green
P ₃	Z	Red



 Sname
 Pname

Catalog

Sid	Pid
S ₁	P ₁
S ₂	P ₂
S ₃	P ₃

$(S.sid = C.sid \wedge C.pid = P.pid \wedge P.color = 'Red')$

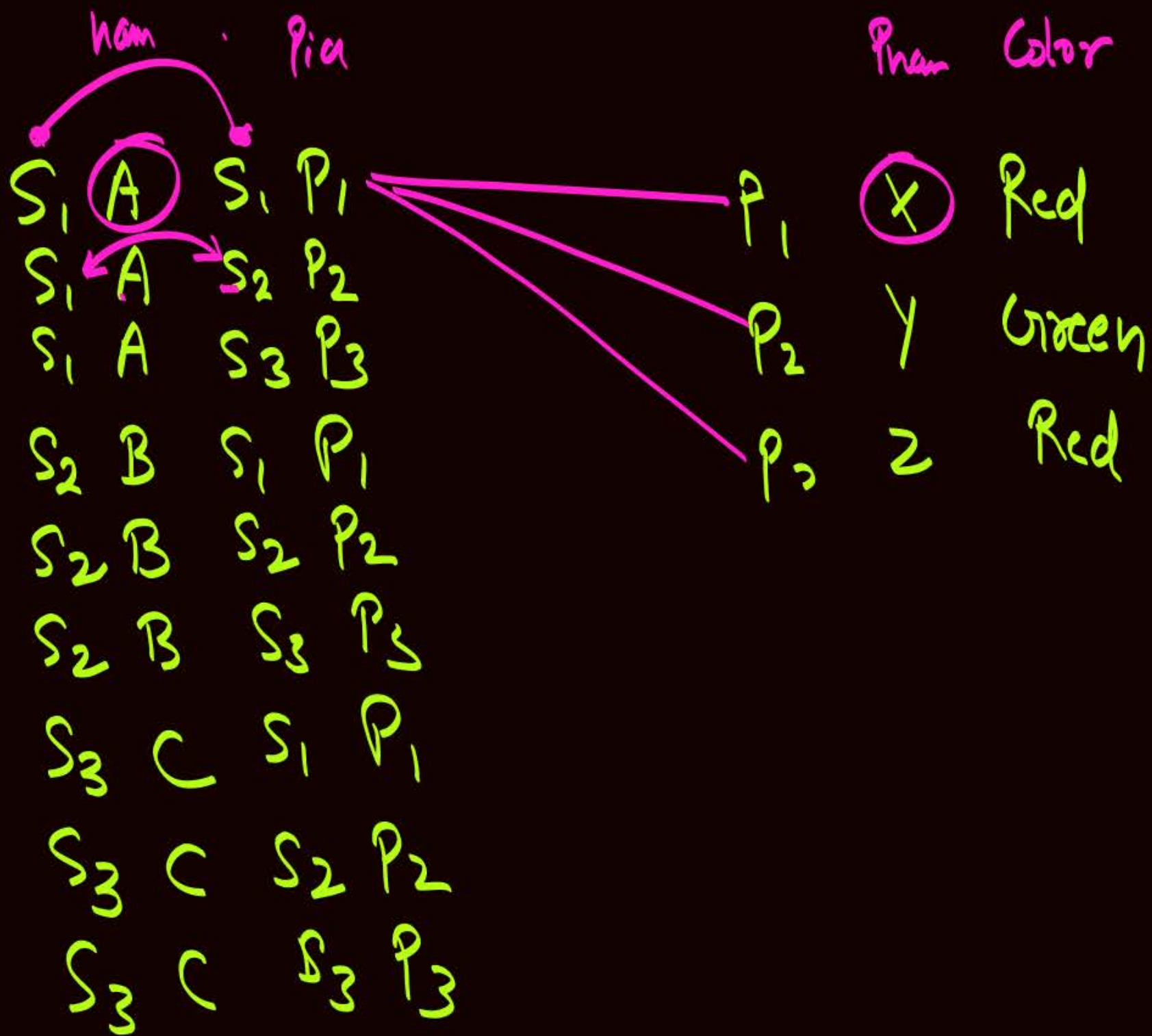
 $(S \times P \times C)$

ham pic

S_1 A S_1 P₁
 S_1 A S_2 P₂
 S_1 A S_3 P₃
 S_2 B S_1 P₁
 S_2 B S_2 P₂
 S_2 B S_3 P₃
 S_3 C S_1 P₁
 S_3 C S_2 P₂
 S_3 C S_3 P₃

Pham Color

X Red
 Y Green
 Z Red



#Q. Consider the following relational schema:

✓ Suppliers(sid, sname, city)

✓ Parts(pid, pname, color)

Catalog(sid, pid, cost)

Find Sname of all the suppliers who have supplied some 'red' color parts along with the names of those parts.

$\{t \mid \exists S \in \text{Suppliers} (\exists C \in \text{Catalog} (C.\text{sid} = S.\text{sid} \wedge \exists P \in \text{Parts} (C.\text{pid} = P.\text{pid} \wedge P.\text{color} = \text{'Red'} \wedge$

$(t.\text{sname} = S.\text{sname} \wedge t.\text{pname} = P.\text{pname})) \}$



Topic : Unsafe TRC query



$\{ t \mid t \notin \text{Relation} \}$

or

$\{ t \mid \neg(t \in \text{Relation}) \}$

Here we are trying to select all the tuples which do not belong to the Relation

A Complement of
"a set of tuples
belonging to relation."

If domain of an attribute happens to be infinite then such query may produce infinite number of tuples in output.

- Unsafe TRC query may produce infinite number of tuples.
- We can never write an Equivalent relational algebra query with respect to an unsafe TRC query.
- For Every safe TRC query we can always write an Equivalent relational algebra query.

Power of TRC query $\neq$ Power of Relational Algebra
(Safe + Unsafe)

Power of Safe TRC query $\equiv$ Power of Relational Algebra

Which of the following relational calculus expressions is not safe?



- A. $\{t | \exists u \in R1 (t[A] = u[A]) \wedge \neg \exists s \in R2 (t[A] = s[A])\}$
- B. $\{t | \forall u \in R1 (u[A] = "x" \Rightarrow \exists s \in R2 (t[A] = s[A] \wedge s[A] = u[A]))\}$
- C. $\{t | \neg(t \in R1)\}$
- D. $\{t | \exists u \in R1 (t[A] = u[A]) \wedge \exists s \in R2 (t[A] = s[A])\}$

Consider the relation employee (name, sex, supervisorName) with name as the key. Supervisor Name gives the name of the supervisor of the employee under consideration. What does the following Tuple Relational Calculus query produce?

$\{e.name \mid \text{employee}(e) \wedge (\forall x) [\neg \text{employee}(x) \vee x.\text{supervisorName} \neq e.name \vee x.\text{sex} = \text{"male"}]\}$

- A. Names of employees with a male supervisor.
- B. Names of employees with no immediate male subordinates
- C. Names of employees with no immediate female subordinates
- D. Names of employees with a female supervisor.

#Q. Which of the following tuple relational calculus expression(s) is/are equivalent to $\forall t \in r(P(t))$?

I $\neg \exists t \in r(P(t))$

II $\exists t \notin r(P(t))$

III $\neg \exists t \in r(\neg P(t))$

IV $\neg \exists t \notin r(\neg P(t))$

A. I only

B. II only

C. III only

D. III and IV only

H.W.
#Q.

Consider The Following Relational Scheme

Student (school-id, sch-roll-no, sname, saddress)

School (school-id, sch-name, sch-address, sch-phone)

Enrolment (school-id, sch-roll-no, erollno, examname)

ExamResult (Erollno, examname, marks)

Consider the following tuple relational calculus query

$\{ t \mid \exists E \in \text{Enrolment } t = E.\text{school-id} \wedge$

$\{ x \mid x \in \text{Enrolment} \wedge x.\text{school-id} = t \wedge (\exists B \in \text{ExamResult } B.\text{erollno} = x.\text{erollno} \wedge$

$B.\text{examname} = x.\text{examname} \wedge B.\text{marks} > 35 \} \mid \div$

$\{ x \mid x \in \text{Enrolment} \wedge x.\text{school-id} = t \} \mid * 100 > 35 \}$

If a student needs to score more than 35 marks to pass an exam what does the query return?

$A = \{1, 2, 3, 4\}$

$|A| = 4$

$|\{1, 2, 3, 4\}| = 4$



#Q.

- A. The empty set
- B. schools with more than 35% of its students enrolled in some exam or the other
- C. schools with a pass percentage above 35% over all exams taken together
- D. schools with a pass percentage above 35% over each exam

H.W.

#Q.



Consider a database that has the relation schemas
EMP(EmpId, EmpName, DeptId) and
DEPT(DeptName, DeptId).

Note that the DeptId can be permitted to a NULL in the relation EMP. Consider the following queries on the database expressed in tuple relational calculus.

- I. $\{t \mid \exists u \in \text{EMP} (t[\text{EMPName}] = u[\text{EmpName}] \wedge \forall v \in \text{DEPT} (t[\text{DeptId}] \neq \text{DeptId}))\}$
- II. $\{t \mid \exists u \in \text{EMP} (t[\text{EMPName}] = u[\text{EmpName}] \wedge \exists v \in \text{DEPT} (t[\text{DeptId}] \neq \text{DeptId}))\}$
- III. $\{t \mid \exists u \in \text{EMP} (t[\text{EMPName}] = u[\text{EmpName}] \wedge \exists v \in \text{DEPT} (t[\text{DeptId}] = \text{DeptId}))\}$

Which of the above queries are safe?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III



2 mins Summary



Topic

Introduction to relational calculus

Topic

Tuple relational calculus

Topic

Practice questions

THANK - YOU